SYNOPSIS

Arogya Mitra: AI-Based Personal Health Tracking and Risk Monitoring System

---

1. Introduction

Arogya Mitra is a web-based AI-powered personal health monitoring platform that allows users to record daily health parameters, analyze trends, and receive probability-based health risk insights using machine learning. The system is intended for preventive health awareness and does not provide medical diagnosis.

---

2. Objective

The main objectives of Arogya Mitra are:
• Provide structured daily health tracking
• Automatically calculate BMI and health score
• Visualize health trends through dashboards
• Generate ML-based risk predictions
• Provide preventive alerts and recommendations
• Ensure secure and privacy-focused data handling
• Maintain scalable modular architecture

---

3. Overview

Arogya Mitra includes secure authentication, health data logging, BMI and health score computation, machine learning risk prediction, interactive dashboards, smart alerts, goal tracking, and downloadable reports. The database is normalized to Third Normal Form (3NF) for consistency and scalability.

---

4. Project Categories

• Artificial Intelligence & Machine Learning Application
• Web-Based Health Monitoring System
• Preventive Healthcare Platform
• Secure Client-Server Application

---

5. Hardware Requirements

Client Side
• Modern web browser
• Internet connection
• Minimum 4GB RAM

Server Side
• Multi-core processor
• Minimum 8GB RAM
• SSD storage
• HTTPS-enabled server

---

6. Software Requirements

Frontend
• React.js
• HTML5, CSS3, JavaScript
• Chart.js / Recharts

Backend
• Python 3.x
• Django
• Django REST Framework
• JWT authentication
• Bcrypt hashing

Machine Learning
• Scikit-learn
• Pandas
• NumPy

Database
• MySQL (3NF normalized)

Security & Deployment
• TLS 1.2+
• Secure environment configuration
• Automated backups

---

7. Structure of the Project

7.1 Presentation Layer

• Login and registration
• Health data entry
• Dashboard and alerts
• Report download

7.2 API Layer

• Authentication APIs
• Health record APIs
• Prediction APIs
• Goal and report APIs

7.3 Service Layer

• Authentication service
• Health analytics
• BMI service
• ML prediction service
• Alert engine
• Report service

7.4 Database Layer

Key tables include: users, user_profiles, health_records, bmi_history, health_scores, risk_predictions, goals, alerts, reports, audit_logs.

---

8. Module Description

8.1 User Management Module

• User registration and login
• Secure password hashing
• Profile management
• Role-based access

8.2 Health Data Logging Module

Users record weight, height, blood pressure, blood sugar, heart rate, sleep hours, water intake, and step count with proper validation and history tracking.

8.3 BMI & Health Score Module

• Automatic BMI calculation
• WHO classification
• Health score (0–100)

8.4 Machine Learning Risk Prediction

• Diabetes and heart disease risk prediction
• Probability scoring
• Risk classification

8.5 Trend Analysis Module

• Weekly and monthly graphs
• Comparative dashboard

8.6 Smart Alert Module

• Abnormal value alerts
• Risk warnings
• Goal notifications

8.7 Goal Tracking Module

• Weight, water, and step goals
• Progress tracking

8.8 Report Generation Module

• Monthly PDF report
• Graphical summaries
• Risk overview

8.9 Privacy & Security Module

• Secure authentication
• Encrypted communication
• Audit logging

---

9. Future Scope

• Mobile application
• Wearable integration
• Personalized AI recommendations
• Multi-language support
• Doctor consultation feature

---

10. Advantages

• Supports preventive healthcare
• AI-based insights
• User-friendly dashboard
• Secure and scalable
• Real-time monitoring

---

11. Limitations

• Not a medical diagnostic system
• Depends on user input accuracy
• Requires internet connectivity
• ML accuracy depends on dataset

---

12. Conclusion

Arogya Mitra is a secure and scalable AI-enabled personal health monitoring platform built using React, Django, and machine learning. It promotes preventive healthcare through continuous monitoring and intelligent analytics while maintaining a strict non-diagnostic approach.